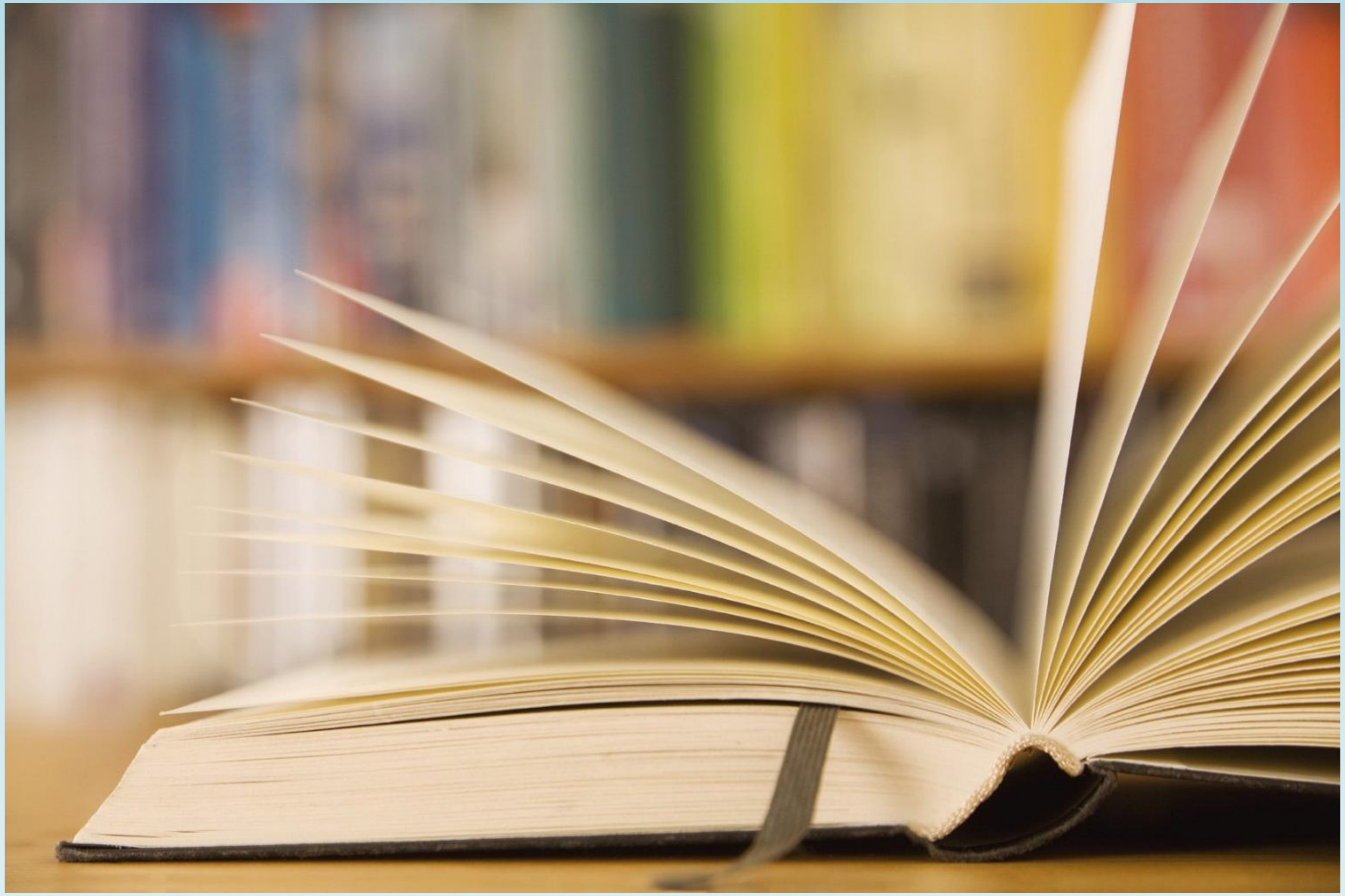


Use Cases

BIS601

Use Case Modelling

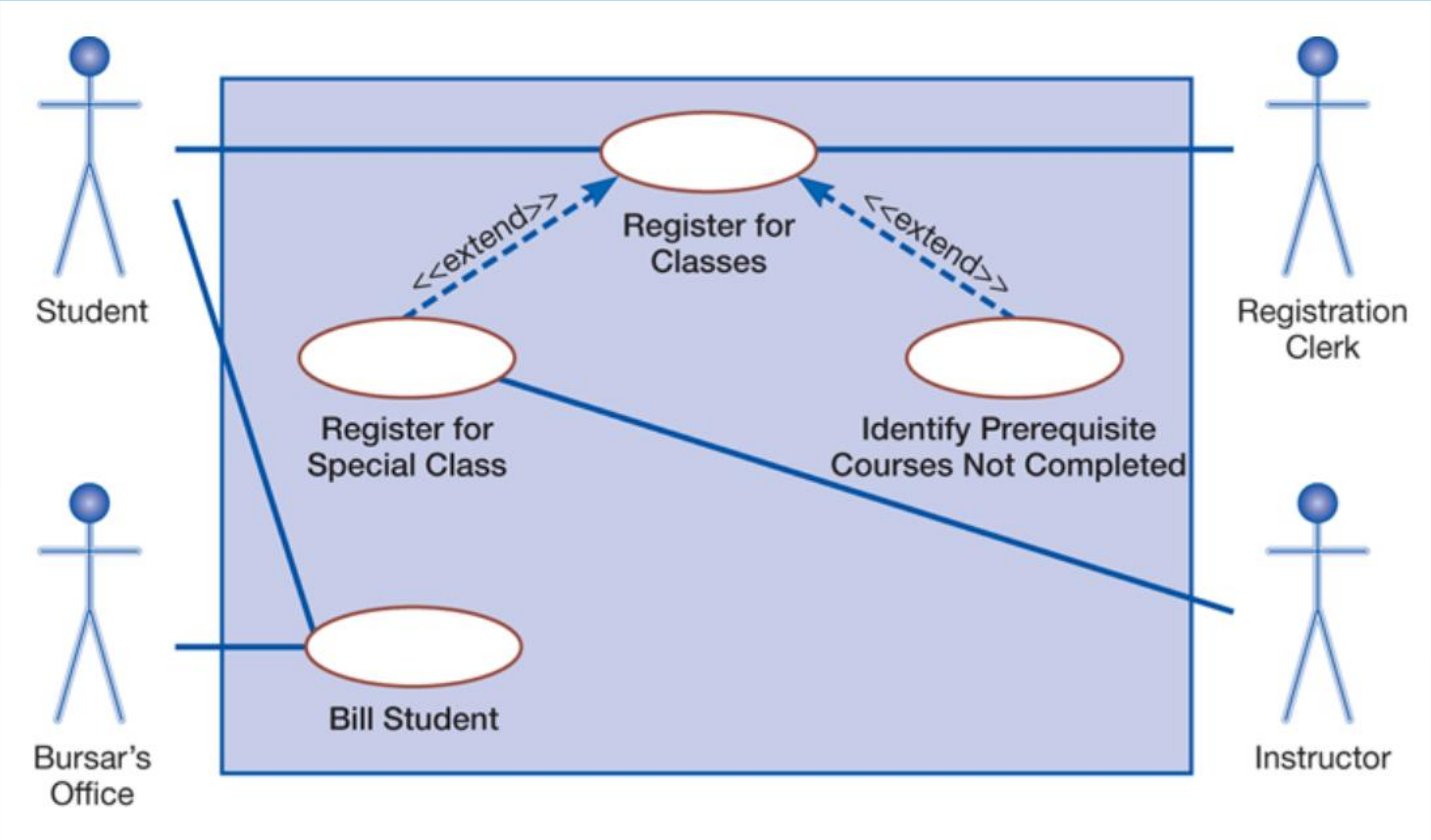
- Use case modelling helps developers understand the functional requirements of the system without worrying about how those requirements will be implemented

Use Case

- Show the behavior or functionality of a system as the system responds to requests from users
- Consist of a set of possible sequences of interactions between a system and a user in a particular environment, possible sequences that are related to a particular goal

Use Case Diagram

- A picture showing system behavior, along with the key actors that interact with the system


Student
Registration Clerk
Register for Classes
<<extend>>
<<extend>>
Register for Special Class
Identify Prerequisite Courses Not Completed
Bill Student
Bursar's Office
Instructor

Symbols

- Actors: A role, not an individual. Stick figure
- Use case: Ellipse. The name of the use case can be listed inside the ellipse or just below it
- System boundary: A box that includes all of the relevant use cases

Symbols

- Connections
  - Actors are connected to use cases with lines
  - Use cases are connected to each other with arrows
  - A solid line connecting an actor to a use case shows that the actor is involved in that particular system function

Symbols

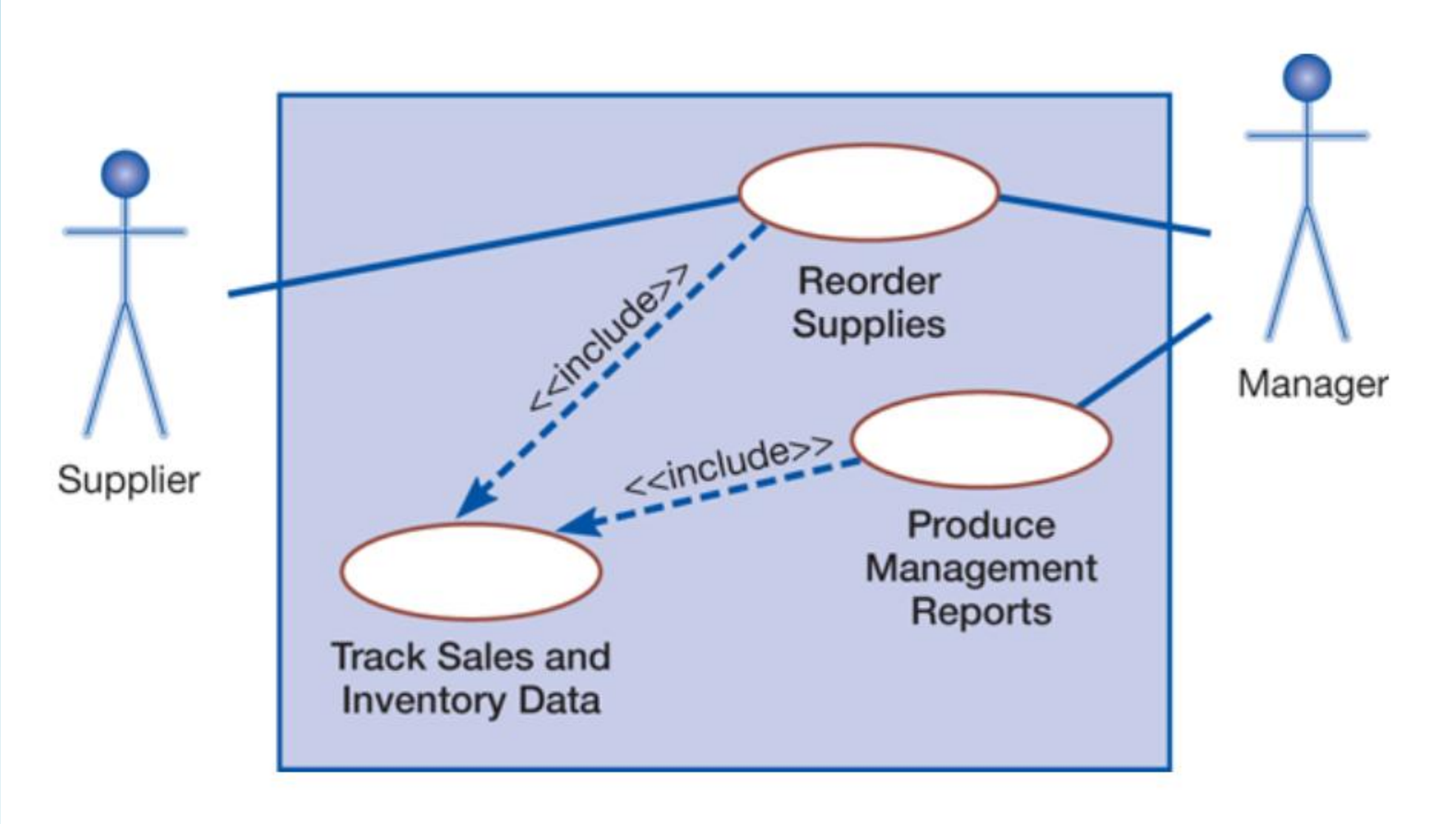
- Extended relationship: Extend a use case by adding new behavior or actions. Shown as a dotted-line arrow pointing toward the use case that has been extended and labelled with the <<extend>> symbol

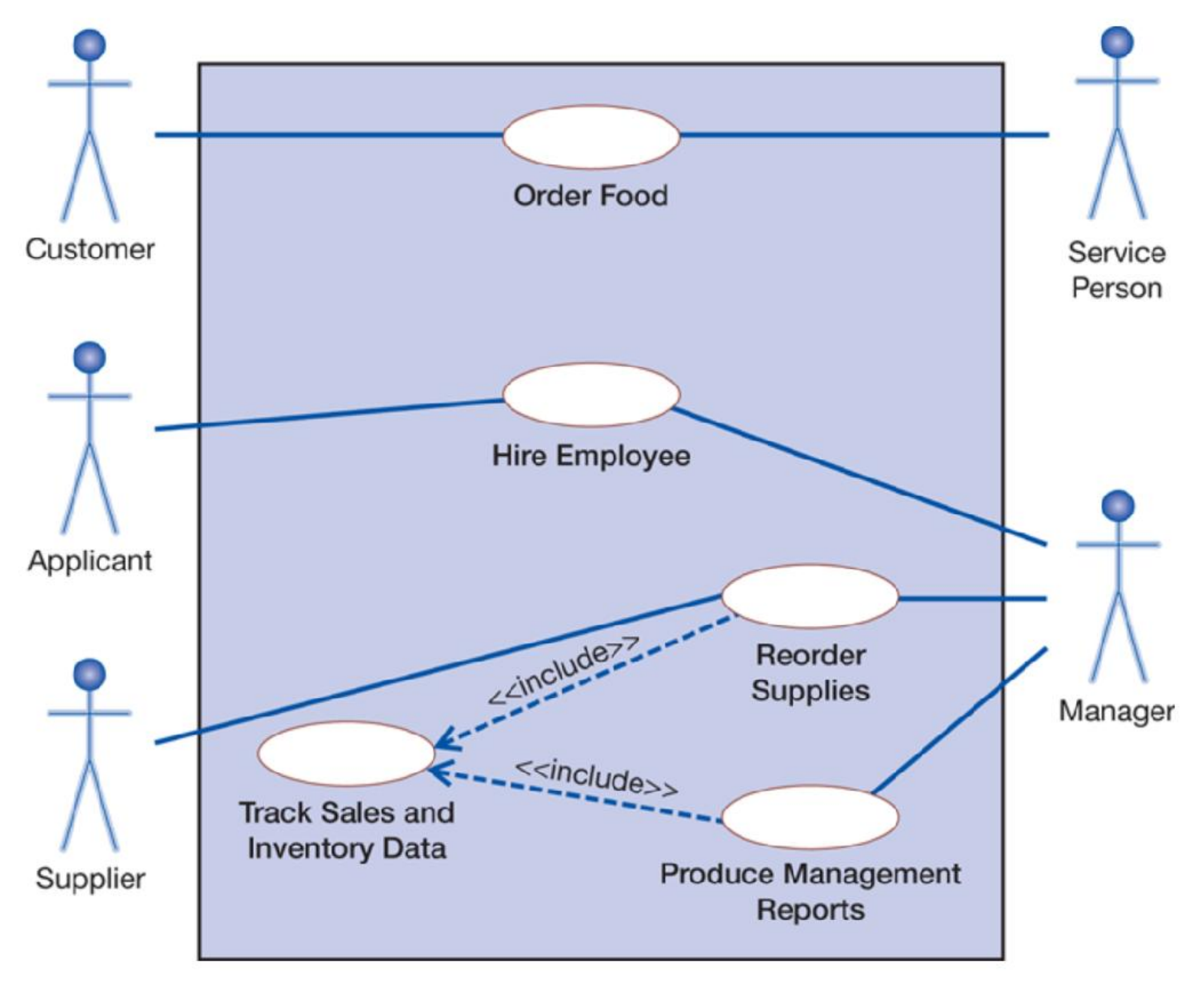
Symbols

- Include relationship: One use case uses another use case. Shown as a dotted-line arrow pointing toward the use case that is being used and labelled with the <<include>> symbol
  - The use case where the arrow originates uses the use case where the arrow ends while it is executing
  - The use case that is “included” represents a generic function that is common to many business functions

Symbols

- Rather than reproduce that functionality within every use case that needs it, the functionality is factored out into a separate use case that can be used by other use cases


Supplier
Reorder Supplies
<<include>>
Manager
<<include>>
Produce Management Reports
Track Sales and Inventory Data


Customer
Order Food
Service Person
Applicant
Hire Employee
Reorder Supplies
Manager
<<include>>
Supplier
Track Sales and Inventory Data
<<include>>
Produce Management Reports

Use Cases

- Explain and document the interaction that is required between the user and the system to accomplish the user's task
- Basis for detailed system functional requirements of business system applications and websites (involving extensive user interactions)
- Helpful in understanding exceptions, special cases, and error handling requirements

Use Case

- A set of activities performed to produce some output results

System:	Displays Welcome message and flashes “Start” button
User:	Presses “Start” button
System:	Displays list of options (buy stamps, mail package, etc.)
User:	Presses “Buy Stamps” button
System:	Displays list of stamp-buying options (type, denomination, quantity)
User:	Selects desired stamps by pressing button next to choice
System:	Displays dollar amount of purchase and asks for purchase confirmation
User:	Presses “Yes” button to confirm purchase
System:	Asks user to swipe card and flashes light next to card reader
User:	Swipes debit card in card reader
System:	Displays keypad for PIN entry and requests PIN
User:	Types in PIN on keypad
System:	Confirms payment transaction and asks if user wants a printed receipt
User:	Presses “Yes” button to request receipt
System:	Prints receipt and asks user to take receipt
User:	Takes receipt
System:	Displays message to wait while it prints the purchased stamps
User:	Takes stamps when printing is done
System:	Displays Thank You Message for 10 seconds, then displays Welcome message

Use Case

- From the user-system dialogue, the postage purchase transaction is accomplished through a coordinated series of prompts, user entries, and system actions
- What the user does to complete the desired task while working with the system
- User’s perspective on the system

The kiosk system will:

- enable the user to begin a transaction
- display a list of purchase transaction options that are available on the kiosk
- enable the user to specify the type of purchase transaction desired
- display a list of options for the purchase type
- enable the user to specify the desired type of postage and quantity desired
- display the amount due for the purchase
- enable the user to confirm the purchase
- enable the user to pay for the purchase with credit/debit card
- complete the credit/debit card transaction securely
- print receipt if requested by user
- print stamps purchased

Sections of a Use Case

Use Case Name: Create preliminary custom drone order	**ID:** UC-6	**Priority:** High
Actor: Customer		
Description: The customer selects and customizes a commercial drone to purchase		
Trigger: Customer wants to purchase a commercial drone		
Type: ☑ External ☐ Temporal		
Preconditions: 1. The customer is authenticated by logging in to his account 2. The Sales System Order Processing application is online		
Normal Course: 1.0 Order a customized drone 1. The customer selects a base model drone from a list of models 2. The system provides availability status for that model (in stock, out of stock) 3. For out of stock status, system displays expected date available a. Customer accepts future availability date; proceed to step 4 b. Customer rejects future availability date; return to step 1 4. The system displays a list of options and upgrades for the selected model 5. The customer selects desired model options and upgrades 6. Preliminary order with cost estimate is created and displayed 7. Customer may return to step 4, confirm order, save for future consideration, or exit without saving 8. Unconfirmed orders are stored in Unconfirmed Custom Order datastore 9. Confirmed orders are saved in Confirmed Custom Order datastore 10. Shop manager is notified of Confirmed Order requiring approval		
Postconditions: 1. Unconfirmed order is stored in Unconfirmed Custom Order datastore 2. Confirmed order is stored in Confirmed Custom Order datastore 3. Shop manager sent notice of Confirmed Order requiring approval		

Sections of a Use Case

- Basic information
  - Use case name
  - Number
  - Description: the use case’s purpose
  - Priority: high, medium, low
  - Actor: a person, another software system, or a hardware device that interacts with the system to achieve a useful goal (user role)

Sections of a Use Case

- Trigger: the event that causes the use case to begin (external trigger, temporal trigger)

- Preconditions: what needs to be accomplished (the state the system must be in) before the use case begins

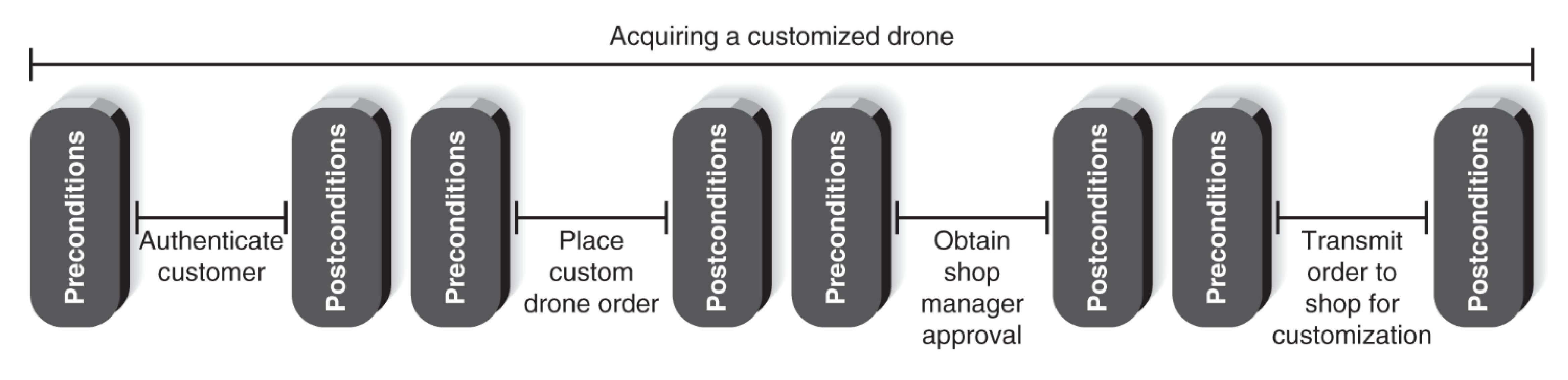
Sections of a Use Case

- Normal course: list the steps that are performed when everything flows smoothly in the system
  - The interactions that occur between the user and the system
  - Listed in the order in which they are performed
  - Branching logical condition (See step 3)

Sections of a Use Case

- Postconditions: the final products of the use case
  - The preconditions of the next use case in the series
- Exceptions: unusual occurrences or errors that may occur as the use case steps are performed

Use Cases in Sequences



Use Cases and Functional Requirements

- The system displays a list of base drone models
- The system accepts customer selection of base drone model
- The system displays in stock/out of stock status for selected drone model
- For an out-of-stock model,
  - The system displays the expected date of availability
  - The system asks customer to accept future date available and continue or to select a different drone model
- The system displays optional features for the selected drone model (batteries, motors, cameras, sensors, etc.)
- The system accepts user choices of options
- The system displays summary and price of selected drone configuration
- The system allows user to continue modifying the drone configuration, save the order for later, confirm the order, or exit without saving.
- For orders saved without confirming, the system stores the order in the Unconfirmed Custom Order datastore
- For confirmed orders,
  - The system displays a completed order summary for the customer
  - The system stores the order in the Confirmed Custom order
  - The system sends a notice of new Confirmed Custom order to the Shop Manager for approval

Creating Use Cases

Step	Activities	Typical Questions Asked[a]
1. Identify the use cases.	Start a use case report form for each use case by filling in the name, description, and trigger.	Ask who, what, when, and where about the use cases (or tasks). What are the major tasks that are performed? What triggers this task? What tells you to perform this task?
2. Identify the major steps within each use case.	For each use case, fill in the major steps needed to complete the task.	Ask how about each use case. What information/forms/reports do you need to perform this task? Who gives you these information/forms/reports? What information/forms/report does this produce and where do they go? How do you produce this report? How do you change the information on the report? How do you process forms? What tools do you use to do this step (e.g., paper, e-mail, phone)?
3. Identify elements within steps.	For each step, identify its triggers and its inputs and outputs.	Ask how about each step. How does the person know when to perform this step? What forms/reports/data does this step produce? What forms/reports/data does this step need? What happens when this form/report/data is not available?
4. Confirm the use case.	For each use case, validate that it is correct and complete.	Ask the user to execute the process, using the written steps in the use case—that is, have the user role-play the use case.

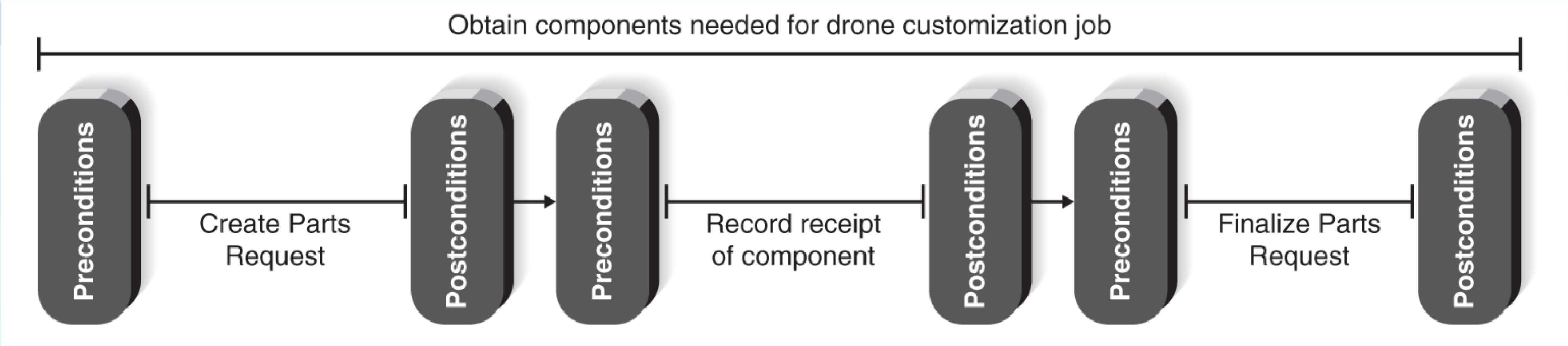
Creating Use Cases

- Event-response list (Drone customization shop management)

Event	Response
1. New shop work order is received	• Shop work order arrival time is recorded
2. Shop work order is assigned to qualified technician	• Work order added to technician's work assignment
3. Parts Request for base model drone and additional components generated	• Parts request stored • Inventory department receives Parts Request
4. Requested parts arrive in Shop Parts room	• Parts arrival recorded on Parts Request
5. All parts on a Parts Request are available	• Technician notified that all parts are available
6. Technician records date/time work begins on work order	• Shop work order updated with start date/time
7. Technician records completion of work on work order	• Shop work order updated with completion date/time • Customer is notified of order completion

Creating Use Cases

- Chain of use cases for obtaining drone components



Creating Use Cases

- Identify the major steps: basic information

Use Case Name: Create Parts Request	**ID:** UC-3	**Priority:** High
Actor: Shop Manager		
Description: This use case describes how the Shop Manager creates a Parts Request.		
Trigger: Shop manager receives notice of new shop work order arrival from Sales System. **Type:** ☑ **External** ☐ **Temporal**		
Preconditions: Shop manager is authenticated Parts Request application is available and online Inventory application is available and online		

Creating Use Cases

Use Case Name: Record Receipt of Component	**ID:** UC-4	**Priority:** High
Actor: Shop Parts Room Clerk		
Description: This use case describes how the Parts Room Clerk Records delivery of drone component from Inventory Department.		
Trigger: Drone component arrives in Parts Room from Inventory Department. **Type:** ☑ **External** ☐ **Temporal**		
Preconditions: Parts Room Clerk is authenticated Parts Request application is available and online		

Creating Use Cases

Use Case Name: Finalize Parts Request	**ID:** UC-5	**Priority:** High
Actor: Parts Room Clerk		
Description: This use case describes how the Parts Room Clerk finalizes a Parts Request.		
Trigger: Notification received that all parts are available for a Parts Request. **Type:** ☑ **External** ☐ **Temporal**		
Preconditions: Parts Room Clerk is authenticated All components listed on Parts Request have arrived in Parts Room		

Creating Use Cases

- Normal course, postconditions

Use Case Name: Create Parts Request	**ID:** UC-3	**Priority:** High

Actor: Shop Manager

Description: This use case describes how the Shop Manager creates a Parts Request

Trigger: Shop Manager receives notice of new shop work order arrival from Sales System

Type: ☑ External ☐ Temporal

Preconditions:
1. Shop manager is authenticated
2. Parts Request application is available and online
3. Inventory application is available and online

Normal Course:	**Information for Steps**
1.0 Create Parts Request 1. Shop manager opens the shop work order 2. Shop manager opens a blank Parts Request 3. For each required component part a. Shop manager enters component part and quantity needed b. System queries the inventory datastore and records part status: (In stock/ Out of stock with expected date available) 4. Shop manager verifies Parts Request is complete 5. System stores new Parts Request 6. System transmits notice of Parts Request to be filled to Inventory Department	

Postconditions:
1. New Parts Request record created and stored
2. Inventory Department notified of Parts Request to be filled

Creating Use Cases

<table>
<tr><td>Use Case Name: Record Receipt of Component</td><td>ID: UC-4</td><td>Priority: High</td></tr>
<tr><td colspan="3">Actor: Shop Parts Room Clerk</td></tr>
<tr><td colspan="3">Description: This use case describes how the Shop Parts Room clerk records delivery of component from Inventory department.</td></tr>
<tr><td colspan="3">Trigger: Drone component arrives in Parts Room from Inventory department</td></tr>
<tr><td colspan="3">Type: ☑ External ☐ Temporal</td></tr>
<tr><td colspan="3">Preconditions:
1. Parts room clerk is authenticated
2. Parts Request application is available and online</td></tr>
<tr><td colspan="2">Normal Course:
1.0 Record Receipt of Component
1. Parts Room clerk retrieves correct Parts Request
2. Parts Room clerk verifies part is correct and undamaged
3. System records date/time part is received
4. Parts Room clerk records the Parts Room location where the part will be stored until pickup
5. System stores updated Parts Request
6. System checks to see if all parts on the Parts Request are available; if they are, the Parts Room clerk is notified to perform Finalize Parts Request use case</td><td>Information for Steps</td></tr>
<tr><td colspan="3">Postconditions:
1. Parts request updated with received part</td></tr>
</table>

Creating Use Cases

Use Case Name: Finalize Parts Request	ID: UC-5	Priority: High
Actor: Shop Parts Room clerk		
Description: This use case describes how the Parts Room clerk finalizes a Parts Request		
Trigger: Notification received that all parts are available for a Parts Request		
Type: ☑ External ☐ Temporal		
Preconditions: 1. Parts room clerk is authenticated 2. All components listed on Parts Request have arrived in Parts Room		
Normal Course: 1.0 Finalize Parts Request 1. Parts room clerk opens the Parts Request and the associated Shop Work Order 2. Parts room clerk verifies that all listed components are currently in the Parts Room 3. Parts room clerk changes the Parts Request Status to "complete" 4. System records the completion date/time in the Shop Work Order 5. System notifies the assigned technician that job is ready	**Information for Steps**	
Postconditions: 1. Parts Request status is complete 2. Date/time all parts are available recorded in Shop Work Order 3. Technician notified that job is ready		

Creating Use Cases

- Identify elements within steps (inputs, outputs)

Use Case Name: Create Parts Request	**ID:** UC-3	**Priority:** High

Actor: Shop Manager

Description: This use case describes how the Shop Manager creates a Parts Request

Trigger: Shop Manager receives notice of new shop work order arrival from Sales System

Type: ☑ External ☐ Temporal

Preconditions:
1. Shop manager is authenticated
2. Parts Request application is available and on-line
3. Inventory application is available and on-line

Normal Course:	**Information for Steps**
1.0 Create Parts Request	
1. Shop manager retrieves the shop work order	→ Shop work order ID ← Shop work order
2. Shop manager opens a blank Parts Request	
3. For each required component part	
a. Shop manager enters component part and quantity needed	→ Part ID, quantity needed
b. System queries the inventory datastore and records part status: (In stock/ Out of stock with expected date available)	← Part status in inventory
4. Shop manager verifies Parts Request is complete	→ Part Request status
5. System stores new Parts Request	→ New Parts Request record
6. System transmits notice of Parts Request to be filled to Inventory Department	→ Parts Request to fill

Postconditions:
1. New Parts Request record created and stored
2. Inventory Department notified of Parts Request to be filled

Summary Inputs	Source	Summary Outputs	Destination
Shop work order ID	Shop manager	Shop work order	Shop manager
Part ID and quantity	Shop manager	Part inventory status	Inventory system
Part request status	Shop manager	New part request record	Parts Request datastore
		Parts Request to fill	Inventory department

Creating Use Cases

Use Case Name: Record Receipt of Component	**ID:** UC-4	**Priority:** High

Actor: Shop Parts Room Clerk

Description: This use case describes how the Shop Parts Room clerk records delivery of component from Inventory department.

Trigger: Drone component arrives in Parts Room from Inventory department

Type: ☑ External ☐ Temporal

Preconditions:
1. Parts room clerk is authenticated
2. Parts Request application is available and on-line

Normal Course:	**Information for Steps**
1.0 Record Receipt of Component	→ Part Request ID
1. Parts Room clerk retrieves correct Parts Request	← Part Request record
2. Parts Room clerk verifies part is correct and undamaged	
a. For incorrect part, enter "Incorrect part–returned" on Delivery slip and return to Inventory department	→ Part return to Inventory Department
b. For damaged part, enter "Damaged part–rejected" on Delivery slip and return to Inventory department	→ Part rejection to Inventory Department
c. Close Part Request and exit use case	
3. System records date/time part is received	→ Date/time received
4. Parts Room clerk records the Parts Room location where the part will be stored until pickup	← Parts room storage location
5. System stores updated Parts Request	→ Updated Parts Request
6. System checks to see if all parts on the Parts Request are available; if they are, the Parts Room clerk is notified to perform Finalize Parts Request use case	→ Part Request Complete notification

Postconditions:
1. Parts request updated with received part
2. Returned/rejected parts sent back to Inventory department

Summary Inputs	Source	Summary Outputs	Destination
Part Request ID	Parts room clerk	Part Request record	Parts room clerk
Parts room storage location	Parts room clerk	Updated Parts Request	Parts Request datastore
		Parts Request Complete notice	Parts room clerk

Creating Use Cases

Use Case Name: Finalize Parts Request	**ID:** UC-5	**Priority:** High

Actor: Shop Parts Room clerk

Description: This use case describes how the Parts Room clerk finalizes a Parts Request

Trigger: Notification received that all parts are available for a Parts Request

Type: ☑ External ☐ Temporal

Preconditions:
1. Parts room clerk is authenticated
2. All components listed on Parts Request have arrived in Parts Room

Normal Course:	**Information for Steps**
1.0 Finalize Parts Request	
1. Parts room clerk opens the Parts Request and the associated Shop Work Order	← Parts Request record ← Shop Work Order record
2. Parts room clerk verifies that all listed components are currently in the Parts Room	
3. Parts room clerk changes the Parts Request Status to "complete"	→ Final parts verification
4. System records the completion date/time in the Shop Work Order	→ Date/time completed
5. System notifies the assigned technician that job is ready	→ Work order ready

Postconditions:
1. Parts Request status is complete
2. Date/time all parts are available recorded in Shop Work Order
3. Technician notified that work order is ready

Summary Inputs	Source	Summary Outputs	Destination
Final parts verification Date/time completion	Parts room clerk Parts room clerk	Parts request record Shop work order record Work order ready notice	Parts room clerk Shop work order datastore Technician